

Mr. Kaveen Weliweriya Liyanage

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Research Interests

Core research interests focus on affective computing and user modelling, generative AI systems and AI security, examining how generative AI can represent and align with the emotional and psychological diversity of human experience.

Education

MSc in Applied Artificial Intelligence | Sep 2025 – Sep 2026

University of Warwick, Coventry, UK

Modules: Artificial Intelligence and Deep Learning, Programming for Artificial Intelligence, Applied Statistics for Artificial Intelligence, Artificial Intelligence and Data Mining, Artificial Intelligence for Industry, Ethical Artificial Intelligence Implementation, MSc Project.

BSc (Hons) in Computing | Jun 2022 – Jun 2024

National Institute of Business Management, Sri Lanka (affiliated with Coventry University), Sri Lanka

Modules: User Experience Designing I & II, Data and Information Retrieval, Programming, Data Structures and Algorithms, Technology and its Social, Legal and Ethical Context, iOS Application Development, Agile Development, Web API Development, Interactive Pervasive Computing, Artificial Intelligence and Human Computer Interaction, BSc Project.

Grade: First Class

Awards and Funding

WMG Excellence Scholarship | Jun 2025

WMG, University of Warwick, Coventry, UK

Merit-based scholarship recognising exceptional academic achievement, leadership potential and professional excellence.

Dean's List Award | Jul 2024

National Institute of Business Management, Sri Lanka, in partnership with Coventry University, United Kingdom

In recognition of the most outstanding academic performance in BSc (Hons) Computing 22.1, Coventry University, United Kingdom.

Publications

Liyanage, K. and Balalle, H. (2024) Emotionally Resonant Branding: The Role of AI in Synthesising Dynamic Brand Images for Artists in the Music Industry. *Open Journal of Applied Sciences*, 14, 2661–2678. doi: [10.4236/ojapps.2024.149175](https://doi.org/10.4236/ojapps.2024.149175). [7 citations]

Research Experience

Affective Sonic Agents for Emotion Alignment | Jun 2026 – Sep 2026

MSc Dissertation, University of Warwick, Coventry, UK

- Developed a multi-agent generative AI framework to measure alignment between brand-intended and audience-perceived emotion in synthetic sonic logos using valence–arousal modelling and OCEAN-based audience agents.
- Designed and conducted controlled computational experiments using synthetic data, statistical analysis and distance-based metrics to evaluate emotional alignment, persona-dependent variation and inter-agent disagreement.

Emotionally Resonant Branding | Jan 2023 – Apr 2024 | Published in the *Open Journal of Applied Sciences*

BSc Dissertation, National Institute of Business Management, Colombo, Sri Lanka / Coventry University, UK

- Developed an AI branding prototype combining facial-emotion recognition, emotion–colour mapping and generative AI to create dynamic brand-image concepts from audience emotional responses.
 - Conducted an empirical study with 191 participants, using survey design and SPSS to evaluate system usefulness, trust, acceptance and ethical concerns; published in the *Open Journal of Applied Sciences*.
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Relevant Skills

ML & Deep Learning — PyTorch, TensorFlow, scikit-learn; CNNs, LSTMs, Transformers. Applied across NLP (text classification, sentiment) and computer vision (facial-emotion recognition, object detection with YOLO/OpenCV).

LLMs & Agents — Fine-tuning, RAG, prompt engineering, multi-agent orchestration and evaluation.

AI Safety & Interpretability — LLM red teaming (Garak), prompt-injection and jailbreak resilience, adversarial robustness testing; SHAP, LIME, Grad-CAM.

Responsible AI — Algorithmic auditing, bias and fairness evaluation, ethical risk assessment, privacy-by-design, AI governance frameworks.

Engineering & MLOps — Python, Docker, Git, MLflow, AWS SageMaker, Azure ML; reproducible experiment pipelines.

Research — Statistics, Quantitative Research, Survey Design, Formulating Research Questions, Hypothesis Development and Testing, Thesis/Dissertation Writing

Projects

Comparative Analysis of Amazon Electronics Reviews | *May 2026*

Developed and benchmarked sentiment classifiers on Amazon Electronics reviews using TF-IDF, Word2Vec and FastText features. Compared Logistic Regression, TextCNN, LSTM and CNN-LSTM models, addressing class imbalance through weighted loss and threshold optimisation. Evaluated models using PR-AUC, macro-F1 and balanced accuracy, with feature selection and model explainability using LIME, SHAP and occlusion sensitivity.

Tech Stack: Python, PyTorch, scikit-learn, Pandas, NumPy, NLP, TF-IDF, FastText, LIME, SHAP.

Diabetes Discovery using Supervised and Unsupervised Learning | *Jan 2026*

Built supervised and unsupervised learning models on the CDC Diabetes Health Indicators Dataset to predict diabetes risk and identify population-risk profiles. Benchmarked Logistic Regression, Random Forest and XGBoost, applied K-means, hierarchical clustering and DBSCAN, and analysed feature importance and model calibration.

Tech Stack: Python, scikit-learn, Pandas, NumPy, XGBoost, LIME, SHAP, Matplotlib.

Languages

English, Sinhala (Native)

References

Available upon request